



Impact of waste rubber powder size and replacement ratio on the compressive strength anisotropy and pore structure of 3D printed concrete

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ABSTRACT

Waste rubber powder, when incorporated into 3D-printed concrete, effectively improves its brittleness but simultaneously alters its compressive strength and pore structure resulting from the low elastic modulus and hydrophobic surface of rubber powder. In particular, the relationship between strength anisotropy induced by the layer-by-layer printing process and compressive strength remains underexplored. Present study explores the impact of rubber powder size and replacement ratio on the buildability and compressive strength anisotropy of 3D-printed concrete. Pore morphology and internal features were examined using mercury intrusion porosimetry (MIP) and scanning electron microscopy (SEM), and their relationships with fractal dimensions, compressive strength, and the anisotropy index (*IA*) were analyzed. Results show that rubber powder significantly enhances flowability; however, excessive incorporation of finer particles may cause segregation, compromising buildability and potentially leading to printing failure. The buildability, characterized by $\tan\theta$, deteriorated most at a 10% rubber replacement ratio, suggesting this threshold should not be exceeded in practice; meanwhile, incorporating 5% of 20-mesh rubber powder effectively minimized compressive strength anisotropy, indicating its potential as an optimal balance between buildability and structural integrity. Furthermore, while the fractal dimension of capillary pores positively correlates with compressive strength in the loading direction (FZ direction), that of macropores shows a strong power-law relationship with *IA*, highlighting its dominant role in the development of strength anisotropy.

1. Introduction

With the continuous advancement of industrialization and intelligent technologies, a growing number of emerging innovations have been introduced into the construction industry [1,2]. Among them, 3D printing technology has attracted widespread attention due to its unique advantages in automation and construction efficiency [3–5]. In particular, 3D-printed concrete significantly reduces

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reliance on traditional formwork, lowers the frequency of formwork removal, decreases labor costs, and effectively conserves construction resources [6]. Moreover, the geometric design of structures can be precisely customized according to specific project requirements, thereby simplifying construction procedures and shortening project timelines. Existing research has systematically explored the flowability, buildability, mechanical, and durability of 3D-printed ordinary concrete [7–11]. Furthermore, this technology has been successfully applied in various practical projects, such as residential buildings [12] and pedestrian bridges [13,14]. Nevertheless, extrusion-based layer-by-layer deposition inevitably introduces interlayer interfaces and pore defects, which can lead to direction-dependent mechanical performance and pronounced compressive anisotropy.

Notably, similar to conventional concrete structures, 3D-printed concrete still suffers from high brittleness and insufficient toughness [15–19]. In conventional cast rubberized concrete, rubber inclusions have been reported to enhance deformability and toughness. However, whether similar mechanisms apply to extrusion-based 3D-printed concrete remains unclear, because the mechanical response of printed materials is strongly governed by printing-induced defects and interlayer interfaces. Fiber reinforcement, such as polypropylene (PP) or polyvinyl alcohol (PVA) fibers, is widely used to enhance toughness through crack-bridging mechanisms [20–23]. However, in extrusion-based 3D printing, fibers may introduce additional processing challenges, for example by affecting extrusion stability and interlayer bonding. Alternatively, viscoelastic inclusions such as rubber particles have been introduced as a toughening strategy [24–26]. Rubberized concrete is a typical example of this strategy. Existing studies have reported that incorporating rubber particles can improve the ductility and crack resistance of conventional cast concrete, although the effectiveness is dependent on rubber size and dosage [27–29]. However, studies on the effects of rubber powder replacement ratio and particle size on the printing process and hardened properties of 3D-printed concrete remain relatively limited, and the underlying trends still need to be systematically clarified.

This is especially true for 3D-printed concrete, where the layer-by-layer stacking process can lead to the formation of void defects between material layers and within the filaments, resulting in significant anisotropy in both the material and structure [30–32]. This anisotropy, caused by weak interlayer interfaces, is one of the key issues currently facing 3D-printed concrete materials and components, and remains a challenge that needs to be addressed [33–35]. For instance, Ding et al. [36] investigated how differences in aggregate properties between natural sand and recycled sand influence the compressive strength and anisotropic behavior of 3D-printed recycled concrete. Existing research has delved into the correlation between microscale pore morphology and the mechanical performance of 3D-printed concrete incorporating different types of aggregates [37–39]. Kruger et al. [40,41] studied the impact of pore structure characteristics within concrete on its strength performance, while Xiao et al. [36,42–44] pointed out that the effects of recycled sand and river sand on 3D-printed concrete properties primarily result from interlayer pore defects and the formation and propagation of initial microcracks within the recycled sand. In addition, compared with natural aggregates, rubber particles differ in hydrophobicity and stiffness [45–47], which may affect material stability and the formation of pore defects during the extrusion-based layer-by-layer deposition process. Accordingly, rubber particle size and dosage are expected to affect buildability and interlayer consolidation during printing, which may in turn reshape pore-defect morphology and thereby influence compressive strength anisotropy rather than inducing a purely uniform strength loss. Therefore, further systematic studies are needed to clarify the above mechanisms and underlying trends.

To evaluate the applicability of rubber powder in 3D-printed concrete, this study first investigates the effects of rubber particle size and dosage on fresh-state buildability and printing quality. Directional compressive tests are then conducted to quantify compressive strength anisotropy under different loading orientations. Mercury intrusion porosimetry (MIP) and scanning electron microscopy (SEM) are further employed to characterize pore structure and microstructural features of mixtures with different rubber parameters, and to reveal the relationships among compressive strength, anisotropy index (IA), pore-defect characteristics, and fractal dimension. Finally, an empirical correlation model linking fractal dimension with compressive strength and IA is established to facilitate mixture evaluation and to provide microstructure-based interpretation for compressive anisotropy in 3D-printed rubberized concrete.

2. Materials and methods

2.1. Raw materials

The cementitious material used was Ordinary Portland Cement (OPC) with a strength grade of 42.5 MPa and an apparent density of 3.10 g/cm^3 , and its chemical composition is detailed in Table 1. The fine aggregates included natural river sand with a particle size range of 0–1.18 mm and an apparent density of 2.64 g/cm^3 , alongside rubber powder. The rubber powder, with particle sizes of 20 mesh, 40 mesh, and 80 mesh as shown in Fig. 1, was used to partially replace the natural fine aggregate in equal volume, with an actual apparent density of 1.1 g/cm^3 . To improve the workability and buildability of the fresh concrete mixture, a polycarboxylate-based superplasticizer (SP) with 25% water-reducing efficiency was incorporated. Fig. 2 illustrates the particle size distribution profiles of the various raw materials used in this study. It should be noted that the particle size of rubber powder was selected based on standard sieve mesh numbers. Given that the typical particle size range of river sand lies between 0 and 1.18 mm, the maximum mesh size of

Table 1
Chemical compositions of cementitious materials used (wt. %).

Oxides	CaO	SiO ₂	Al ₂ O ₃	MgO	SO ₃	Fe ₂ O ₃	K ₂ O	TiO ₂	Others	LOI
Cement	57.40	21.91	6.41	4.39	3.06	2.68	0.90	0.32	0.50	2.43

* Others: MnO, SrO, etc.; LOI: Loss on ignition.

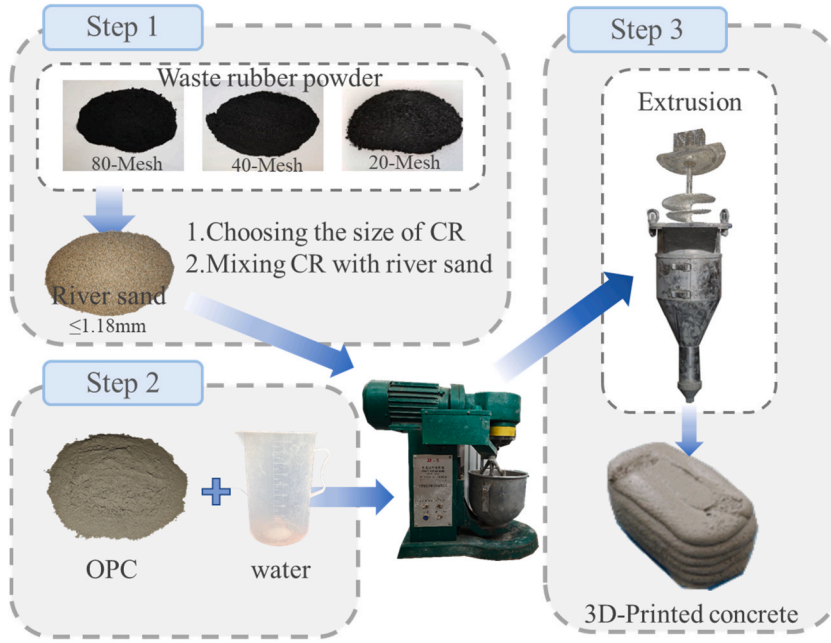


Fig. 1. Raw materials and preparation process for 3D-printed concrete.

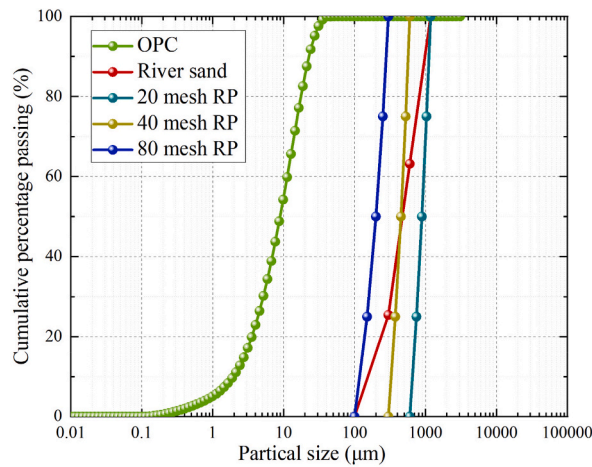


Fig. 2. Particle size distribution curves for all raw materials.

rubber powder was set to 20 mesh, corresponding to a sieve aperture of approximately 0.84 mm, to match the particle scale of natural sand. The particle size was progressively decreased based on mesh numbers, with the smallest being 80 mesh, corresponding to an aperture of approximately 0.18 mm, to explore the effect of finer rubber particles on the performance of 3D-printed concrete.

2.2. Mixture design

As detailed in Table 2, eleven groups of mixtures were formulated across four series by varying rubber powder particle size (three levels) and replacement ratio (four levels), including one control group without rubber. Rubber powder sizes were chosen as 20 mesh, 40 mesh, and 80 mesh based on different sieves, with substitution rates set at 5%, 10%, 15%, and 20%. It should be noted that for the mixture containing 80-mesh rubber powder, the replacement ratios were set only at 5% and 10%. This is because, when the replacement ratio exceeds 10% with 80-mesh rubber powder, the mixture no longer possesses buildability, and therefore, corresponding tests were not conducted. The water-to-cement ratio was designed as 0.35 and SP dosage was designed as 0.08% of cement dosage according to the pre-experimental results to obtain better workability.

Table 2Mix proportions (kg/m³).

Series	Group	W/C	Cement	Water	Sand	CR	SP (%)
O	NC	0.35	953.72	333.80	953.72	-	0.08
I	RPC-20-5%	0.35	953.72	333.80	906.04	19.36	0.08
	RPC-20-10%	0.35	953.72	333.80	858.35	38.72	0.08
	RPC-20-15%	0.35	953.72	333.80	810.67	58.08	0.08
	RPC-20-20%	0.35	953.72	333.80	762.98	77.45	0.08
II	RPC-40-5%	0.35	953.72	333.80	906.04	19.36	0.08
	RPC-40-10%	0.35	953.72	333.80	858.35	38.72	0.08
	RPC-40-15%	0.35	953.72	333.80	810.67	58.08	0.08
	RPC-40-20%	0.35	953.72	333.80	762.98	77.45	0.08
III	RPC-80-5%	0.35	953.72	333.80	906.04	19.36	0.08
	RPC-80-10%	0.35	953.72	333.80	858.35	38.72	0.08

2.3. Specimen preparation

Specimens were fabricated using a laboratory 3D printer, as depicted in Fig. 3. The printer utilizes a Cartesian coordinate system and a truss-frame structure, offering a build area of 1.75 m (X) × 0.7 m (Y) × 1.3 m (Z). The printer is equipped with a 20 mm diameter circular nozzle and operates at a printing speed of 30 mm/s, as shown in Fig. 3.

For each mixture, three cast and three 3D-printed concrete specimens were produced. The cast-in-place specimens were cubes with a side length of 40 mm, while the 3D-printed concrete specimens were rectangular, measuring 150 mm (X) × 60 mm (Y) × 60 mm (Z), as shown in Fig. 4. The 3D-printed specimens for compressive strength test were cut with a saw to match the geometry of the cast specimens (40 mm cubes). Given the three orthogonal loading directions (X, Y, and Z), three cubes per direction were extracted from the same printed element, and each specimen was labeled according to its intended loading direction to prevent mix-ups. All specimens underwent standard curing for 28 days prior to compressive strength testing. Following the curing period, the 3D-printed samples were sectioned to prepare representative specimens for MIP and SEM analysis aimed at microstructural characterization.

2.4. Test procedure

The 3D-printed concrete was first tested for fluidity and buildability prior to strength testing. Fluidity was tested according to GB 2419 [48], with measurements taken immediately after the concrete mixing process was completed. Buildability was characterized using the $\tan\theta$ value [49]. The $\tan\theta$ value, as illustrated in Fig. 5, serves as a comprehensive indicator of the buildability of 3D-printed concrete, where lower values correspond to better buildability. The bottom width W_b and the top width W_t of each specimen were measured to determine the lateral deformation a (e.g., $a = (W_b - W_t)/2$, as defined in this study). The specimen height h was also measured. The buildability index was then calculated as $\tan\theta = a/h$. Considering the irregular edges caused by the printing process, ten measurement points were taken and averaged, and the averaged values of a and h were used to reduce measurement uncertainty.

The strength tests were conducted using a 100-ton MTS press with a loading rate set to 1 kN/s. For each group and for each of the three directions (X, Y, Z), the reported value is the mean of three repeated measurements. To further interpret the differences in compressive strength among the mixtures, MIP and SEM were employed to analyze the pore structure and microstructural features of each group individually. Samples of approximately 1 cm³ were taken from the center of the 3D-printed concrete cubes (following previous research [50–52]), soaked in anhydrous ethanol for 48 h to stop the hydration process, and subsequently dried in a vacuum oven at 60 °C.

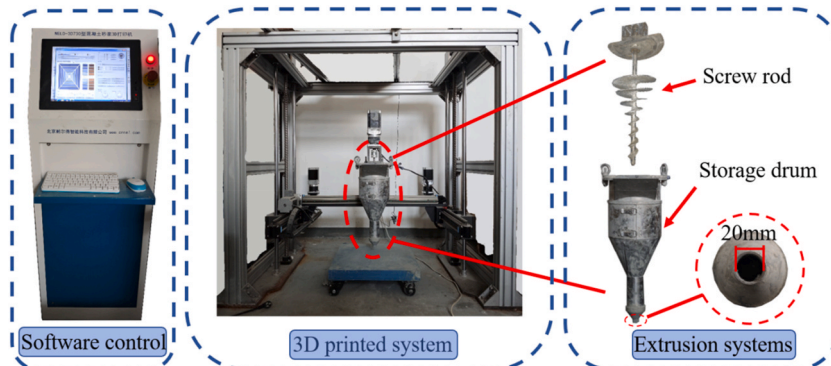


Fig. 3. 3D printed system.

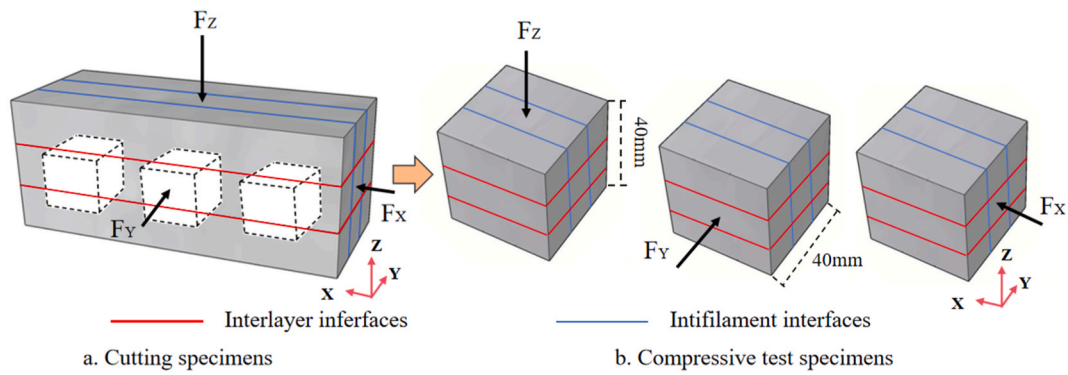


Fig. 4. Dimensions of specimens for mechanical and microstructural testing.

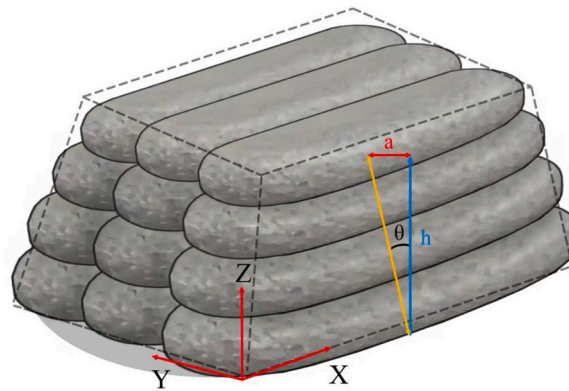
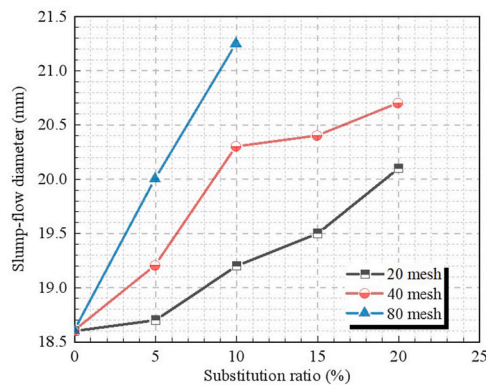


Fig. 5. Testing and calculation of $\tan\theta$ value.

3. Results and analysis

3.1. Slump-flow diameter

Effect of incorporating rubber powder of varying mesh sizes and volume fractions into concrete on slump-flow diameter is plotted in Fig. 6. The slump-flow diameter exhibited a consistent increase with the increment of rubber powder volume fraction. Specifically, in the case of 20-mesh rubber powder, the initial slump-flow diameter of plain concrete was 18.6 mm, which increased to 18.7 mm, 19.2 mm, 19.5 mm, and 20.1 mm at replacement levels of 5%, 10%, 15%, and 20%, respectively. Similar patterns were observed for



(a)

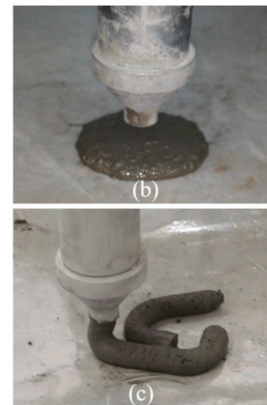


Fig. 6. Slump-flow diameter for each mixture. (a). test results; (b). flows easily and directly; (c). extrusion failure.

rubber powder with 40-mesh and 80-mesh sizes. This can be explained by the hydrophobic properties of rubber powder, which lead to the formation of a distinct interfacial transition zone (ITZ) between the rubber powder and the surrounding mortar. Consequently, the thickness of the paste film between natural aggregates is increased, thereby reducing inter-particle friction and significantly enhancing the slump-flow diameter. Moreover, with the increase in the mesh size of rubber powder, the slump-flow diameter of the concrete increases significantly under the same replacement ratio. For instance, when the replacement ratios are 5% and 10%, the incorporation of 40-mesh rubber powder noticeably enhances the slump-flow diameter, and this effect becomes even more pronounced as the mesh size increases to 80 mesh. It should be noted that when the replacement level exceeded 10%, rubber concrete exhibited pronounced segregation, and the measured flow spread exceeded the measurable maximum value. Therefore, when rubber powder with a mesh size greater than 80 are employed, it is recommended that their incorporation rate should not exceed 10%. The continuous increase in slump-flow with increasing mesh size is likely because finer rubber powder corresponds to smaller particle diameters, resulting in a greater number of dispersed rubber particles within the concrete at the same replacement ratio. The increased dispersion of rubber particles could cause the paste film thickness to exceed a critical limit, ultimately leading to segregation.

Notley, the fresh properties of 3D printed rubber powder concrete must be maintained within an appropriate range to ensure successful extrusion. Experimental results indicated that when the slump-flow diameter ranged between 18 mm and 22 mm, the material could be smoothly extruded through the rotating screw mechanism. However, when the slump-flow diameter exceeded 22 mm, the concrete tended to flow uncontrollably out of the nozzle, as illustrated in Fig. 6b. For instance, in the case of 80-mesh rubber powder, when the replacement ratios were 15% and 20%, the slump-flow diameter surpassed 22 mm. Therefore, these two mixtures were excluded from further experimental investigation. Conversely, when the flow diameter was less than 16 mm (based on preliminary experiments), the concrete lacked sufficient fluidity to pass through the rotating screw, thereby resulting in printing failure, as shown in Fig. 6c.

3.2. Buildability

Buildability is a critical performance indicator for evaluating the ability of 3D-printed concrete to be smoothly extruded, maintain its printed geometry, and adequately support subsequent layers during the printing process. The $\tan\theta$ value, which reflects the horizontal displacement of the top layer during printing, is commonly used to assess buildability [49], as illustrated in Fig. 7. In general, a smaller $\tan\theta$ value indicates better buildability. Buildability is primarily governed by fresh-state rheological parameters, such as static yield stress [53–55], thixotropic structural build-up, and the rate of strength recovery after extrusion. In this context, $\tan\theta$ can be viewed as a geometric manifestation of insufficient fresh-state stability, where larger lateral spreading reflects weaker resistance to gravity-induced deformation during layer-by-layer deposition.

As shown in Fig. 7, the incorporation of rubber powder leads to a progressive increase in $\tan\theta$ compared to plain concrete, indicating a deterioration in buildability. Moreover, the $\tan\theta$ value initially increases and then decreases with increasing rubber powder replacement ratio, reaching a maximum at a 10% replacement level. This is primarily because, at low rubber content (e.g., 5%), the low apparent density and elastic modulus of rubber particles weaken the interlayer support capacity, reducing sidewall stability and thus degrading buildability. When the replacement level increases to 10%, this weakening effect becomes most pronounced, resulting in the highest $\tan\theta$ value. As the replacement ratio continues to rise, more rubber particles (particularly 20-mesh and 40-mesh) are dispersed throughout the matrix. At this stage, the reduced apparent density of rubber particles helps lower the overall weight of the concrete. Additionally, due to the ITZs formed, more cement particles are retained within the slurry, thereby increasing the viscosity of the mix. This enhanced viscosity may improve the interlayer support, contributing to the observed improvement in buildability at higher replacement ratios. From a microstructural perspective, a larger $\tan\theta$ implies more filament spreading and non-uniform layer thickness, which can hinder effective interlayer consolidation and promote printing-induced pore defects (e.g., interlayer voids and

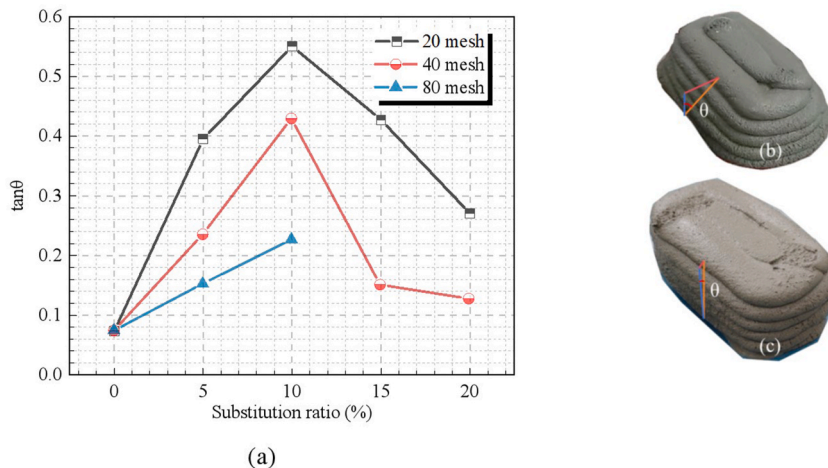


Fig. 7. Buildability of 3D-printed concrete with and without rubber powder. (a). test results for each mixture; (b). high $\tan\theta$; (c). good $\tan\theta$.

interfilament pores). These defect characteristics provide a direct pathway for strength reduction and may also amplify directional disparity in compressive response.

In addition, as the rubber powder particle size decreases, the $\tan\theta$ value shows a declining trend. As illustrated in Fig. 7, under the same replacement ratio, the incorporation of 20-mesh rubber results in the highest $\tan\theta$ value, followed by 40-mesh, while 80-mesh rubber exhibits the lowest value. The primary reason is that finer rubber powder, due to its smaller particle size, occupies a larger volume at the same replacement ratio and is more uniformly distributed throughout the mix. This enhances the encapsulation of cementitious particles and increases the static shear yield strength of the material. Furthermore, the larger specific surface area of finer particles helps bind more free water, promoting early structural build-up and improving interlayer support and bond stability, thereby leading to better buildability (i.e., lower $\tan\theta$ values).

3.3. Compressive strength

Owing to the distinct fabrication techniques of 3D-printed and cast specimens, the compressive strength of printed specimens displays directional dependence under various loading orientations. The compressive strength of 3D-printed concrete cast specimens under both identical and different loading directions are plotted in Figs. 8 and 9, respectively. Fig. 8 first compares the compressive strength of 3D-printed rubberized concrete and cast-in-place concrete under the same loading direction (FZ direction). Fig. 8 shows that as the rubber powder content increases, the compressive strength of cast concrete decreases. For instance, at a 5% replacement with 20-mesh rubber powder, the compressive strength was 37.1 MPa, which decreased to 20.5 MPa when the replacement ratio increased to 20%. A similar trend occurred with 40- and 80-mesh rubber. This reduction is attributed to the low elastic modulus and non-polar nature of rubber, which increase ITZs and weaken the matrix. Additionally, smaller rubber sizes lead to more particles at the same dosage, generating more ITZs and further strength loss. For 3D-printed rubberized concrete specimens, the compressive strength exhibited a similar decreasing trend to that of the cast specimens with increasing rubber powder size and replacement ratio, particularly in the FZ direction. The compressive strength in the FZ direction of 3D-printed rubberized concrete may also be affected by buildability. For instance, the FZ-direction compressive strength of the RPC-20-10% specimen was lower than that of the RPC-40-10%, likely due to a higher $\tan\theta$ value that reduced the interlayer bonding strength.

Furthermore, due to the layer-by-layer nature of the printing process, both interlayer and interfilament interfaces are inevitably formed, resulting in potential weak interfaces. These defects lead to significant differences in compressive strength under different loading directions for 3D-printed concrete. As shown in Fig. 9, compared with cast concrete specimens, the compressive strengths along the FX direction (perpendicular to the printing path) and the FY direction (perpendicular to the filament stripes) also show noticeable reductions. This phenomenon is closely related to the increased porosity caused by the interlayer and interfilament interfaces. Moreover, with the increasing volume replacement of rubber particles with low elastic modulus, the compressive strength also exhibits a decreasing trend. For example, when 20-mesh rubber powder is used, the compressive strength of 3D-printed concrete decreases from approximately 29.0 MPa to 21.4 MPa at a 5% replacement ratio, and further declines to 17.5 MPa and 12.4 MPa at a 20% replacement ratio. A similar decreasing trend in compressive strength is also observed as the rubber powder size decreases.

For 3D-printed plain concrete specimens, the highest compressive strength of 44.6 MPa was observed in the FY direction, followed by 38.1 MPa in the FX direction, and the lowest of 33.4 MPa in the FZ direction. The primary reason is that when the loading direction is perpendicular to the filament bands (FY direction), the overlapping between adjacent filaments and the secondary compaction at the edges generate effective mechanical interlocking, thereby resulting in higher compressive strength. In contrast, when the loading direction is aligned with the printing path (FX direction), under-extrusion channels are more likely to form along the same direction during the extrusion process, reducing the effective load-bearing area and thus leading to lower strength than in the FY direction.

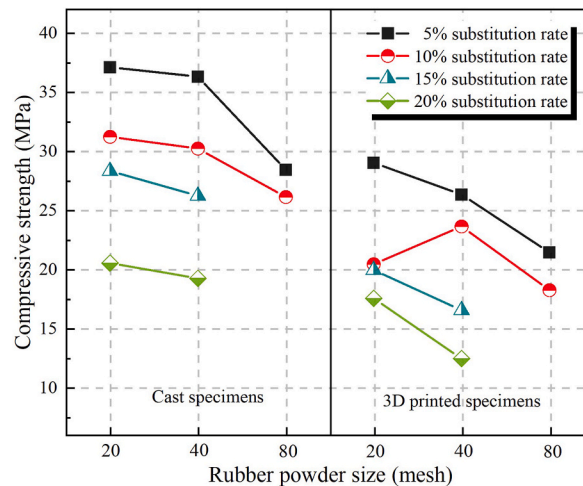


Fig. 8. Compressive strength of cast and 3D-printed concrete specimens in the FZ direction.

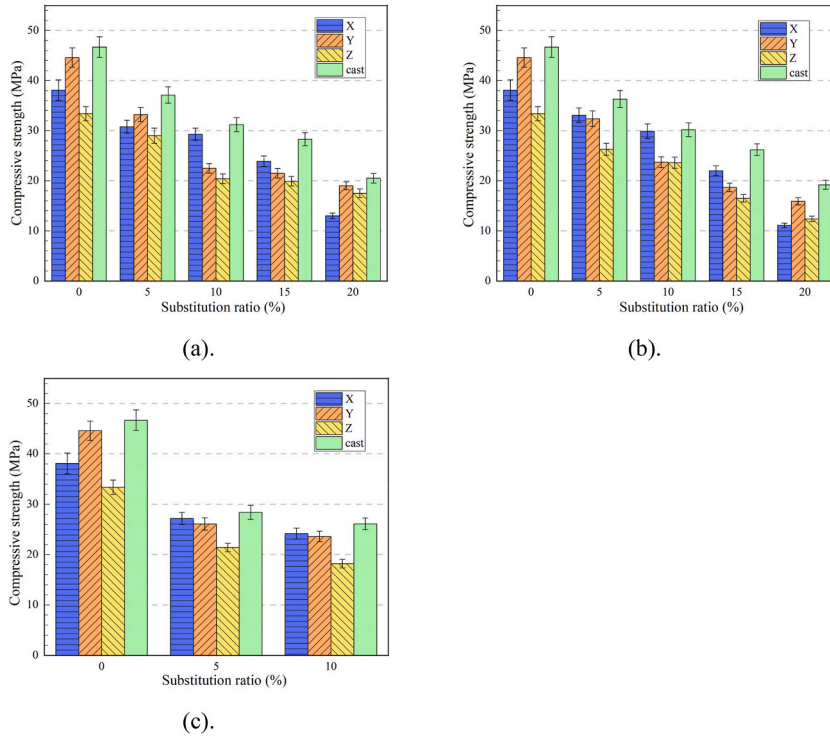


Fig. 9. Compressive strength of 3D-printed concrete and cast-in-place specimens in different loading directions. (a) rubber powder size of 20-mesh; (b) rubber powder size of 40-mesh; (c) rubber powder size of 80-mesh.

Additionally, for the FZ direction, the relatively long deposition intervals between layers result in insufficient interlayer bonding, making interfacial delamination and through-thickness cracking more prone to occur under load. Consequently, the compressive strength in the FZ direction exhibits the lowest value. After incorporating 5% of 20-mesh rubber powder, the variation in compressive strength of 3D-printed concrete specimens across different loading directions remained consistent with that of plain concrete specimens; however, the absolute strength values exhibited a slight reduction. With the increase in replacement ratio, the compressive strength in the FX direction gradually became the highest, followed by FY and then FZ. A similar trend was also observed as the rubber powder particle size decreased to 40 mesh and 80 mesh. The main reason may be that the 20-mesh rubber particles have a relatively large particle size. Therefore, at a 5% replacement ratio, the number of particles dispersed within the 3D-printed concrete is relatively low, and the resulting porosity has a limited effect on the compressive strength in all loading directions. As the replacement ratio increases and the particle size decreases, the volumetric content of rubber powder dispersed within the concrete increases. This leads to the formation of more ITZs around rubber particles, thereby weakening the performance of interlayer and interfilament interfaces. Consequently, the loading directions that must traverse these interfaces (i.e., FY and FZ) suffer more pronounced strength degradation. When the replacement ratio reached 20%, the compressive strength in the FX direction of 3D-printed concrete specimens containing 20-mesh and 40-mesh rubber powder significantly decreased, becoming the lowest among the three loading directions. This may be attributed to the excessive replacement ratio leading to a substantial deterioration in bonding between filaments and layers, which caused premature failure of the specimens and a marked reduction in strength. In contrast, the loading surfaces in the FZ and FY directions involve only interlayer or interfilament interfaces, whereas the FX direction involves both, making it more vulnerable.

3.4. Evaluation of compressive-anisotropic behavior

To quantitatively describe the anisotropy of compressive strength in 3D-printed rubber concrete, existing studies have introduced the anisotropy coefficient as a characterization parameter, as shown in Eqs (1)–(4) [56,57].

$$I_X = \frac{\sqrt{(F_X - F_X)^2 + (F_X - F_Y)^2 + (F_X - F_Z)^2}}{F_X} \quad (1)$$

$$I_Y = \frac{\sqrt{(F_Y - F_X)^2 + (F_Y - F_Y)^2 + (F_Y - F_Z)^2}}{F_Y} \quad (2)$$

$$I_Z = \frac{\sqrt{(F_Z - F_X)^2 + (F_Z - F_Y)^2 + (F_Z - F_Z)^2}}{F_Z} \quad (3)$$

$$I_A = \frac{I_X + I_Y + I_Z}{3} \quad (4)$$

where I_X , I_Y , I_Z represent the anisotropy coefficient in each of the three directions X, Y, Z respectively, and I_A represents the overall anisotropy coefficient.

As shown in Table 3, the I_A values for 3D-printed plain concrete and rubber powder specimens are presented. Theoretically, I_A ranges from 0 to positive infinity. A higher I_A value indicates a more significant anisotropic effect on compressive strength caused by the layer-by-layer printing process and the incorporation of rubber powder. When $I_A = 0$, it suggests that the compressive strength is unaffected by either the printing process or the incorporation of rubber powder. Therefore, investigating the influence of rubber powder size and replacement ratio on I_A is beneficial for optimizing the material formulation and structural performance of 3D-printed rubber powder concrete.

As depicted in Fig. 10, the particle size and replacement ratio of rubber powder considerably affect the I_A of the compressive strength in 3D-printed concrete. For plain concrete, the I_A value is 0.288. After adding 5% rubber powder, the I_A decreases, with the most pronounced reduction observed for the 20-mesh mixture, where I_A drops from 0.288 to 0.134. When the rubber powder size is reduced to 40 mesh and 80 mesh, the I_A values become only slightly lower than that of plain concrete, indicating a relatively limited change in compressive anisotropy. This can be explained by the fact that, at the same replacement ratio, the 20-mesh mixture contains the fewest rubber particles dispersed in the mortar. With fewer inclusions, the hydrophobic surface characteristics of the 20-mesh rubber may promote paste redistribution and facilitate the filling of printing-induced voids, thereby enhancing interlayer and inter-filament bonding. Similar observations have also been reported by Ren et al. Such bonding improvement reduces the disparity in compressive strength among different loading directions, leading to a pronounced decline in I_A . In contrast, as the rubber particle size further decreases to 40 mesh and 80 mesh, the number of dispersed particles increases markedly, and the associated increase in total interfacial area and potential interface weakening becomes a dominant factor affecting compressive anisotropy. However, at the 5% replacement ratio, the effectiveness of these finer particles in improving interfacial bonding diminishes, resulting in higher I_A values than those observed for specimens containing 20-mesh rubber powder.

Moreover, as the rubber powder replacement ratio increases, a greater number of rubber particles are dispersed within the interlayer and interfilament regions of the 3D-printed concrete, which further deteriorates the bonding performance at these interfaces. Consequently, the I_A exhibits a gradually increasing trend. Experimental results show that with higher rubber powder replacement ratios, particularly at 20%, the anisotropy of compressive strength becomes more pronounced, resulting in a significantly higher I_A value compared to plain 3D-printed concrete. Notably, in the case of the 20-mesh rubber powder mixture, the I_A value at a 10% replacement ratio is significantly higher than those at 5% and 15%. This may be closely related to the relatively high $\tan\theta$ value observed for this mixture. A higher $\tan\theta$ indicates poor buildability, which can lead to deformation of printed filaments and inconsistent layer thickness-factors that are likely to exacerbate the anisotropy in compressive strength. For 3D-printed rubberized concrete specimens with 20-mesh and 40-mesh rubber powder, I_A values exceed those of plain concrete at the 20% replacement ratio. Similarly, for 80-mesh rubber powder, I_A shows a similar trend when the replacement ratio is 10%. Experimental results show that in 3D-printed rubberized concrete, finer rubber particles (e.g., 80-mesh) combined with higher replacement ratios significantly increase the anisotropy of compressive strength. Therefore, the replacement ratio should be kept low when using finer rubber powders. In contrast, using relatively larger rubber particles (e.g., 20-mesh) at lower replacement levels can help reduce the anisotropy index and enhance the uniformity of mechanical performance.

Table 3
Anisotropy index calculation for each mixture.

Series	Group	Compressive strength (MPa)				Anisotropy index			
		FX	FY	FZ	Cast	IX	IY	IZ	I_A
O	NC	38.1	44.6	33.4	46.7	0.211	0.290	0.364	0.288
I	RPC-20-5%	30.8	33.2	29	37.1	0.097	0.146	0.158	0.134
	RPC-20-10%	26.3	22.5	20.4	31.2	0.267	0.193	0.307	0.256
	RPC-20-15%	23.9	21.5	19.9	28.3	0.195	0.134	0.216	0.182
	RPC-20-20%	13	19	17.5	20.5	0.577	0.326	0.271	0.391
II	RPC-40-5%	33.1	32.4	26.3	36.3	0.207	0.190	0.347	0.248
	RPC-40-10%	29.9	23.7	23.6	30.2	0.296	0.262	0.267	0.275
	RPC-40-15%	22	18.7	16.5	26.2	0.292	0.212	0.359	0.288
	RPC-40-20%	11.1	15.9	12.4	19.2	0.448	0.374	0.301	0.374
III	RPC-80-5%	27.2	26.1	21.4	28.4	0.217	0.185	0.349	0.250
	RPC-80-10%	24.2	23.6	18.2	26.1	0.249	0.230	0.444	0.308

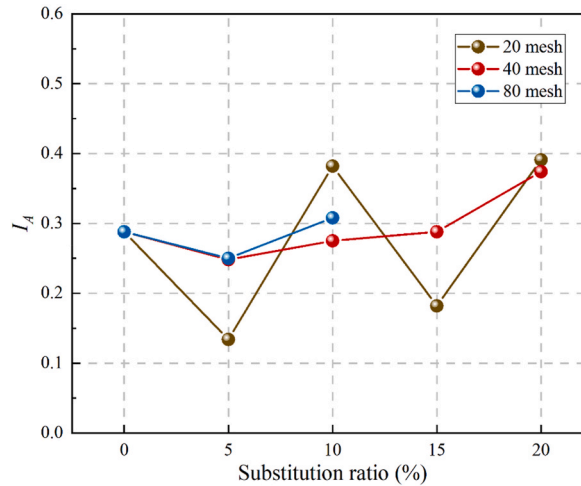


Fig. 10. I_A for 3D-printed concrete specimens with and without rubber powder.

3.5. Micro-morphology

Fig. 11 initially presents the influence of rubber powder particle size variation on the microstructure under a constant replacement ratio. In general, the cast specimens show relatively compact interfaces between the rubber powder and surrounding mortar, with cement paste uniformly distributed around the rubber powder. At the same magnification, no obvious interfacial cracks or debonding phenomena were observed, indicating that the vibration process during casting enhances interfacial bonding and results in a denser internal structure. In contrast, the 3D-printed specimens display noticeable interfacial defects between rubber particles and adjacent mortar. Distinct ITZs can be observed, which may be one of the contributing factors to the inferior mechanical properties of 3D-printed specimens relative to cast specimens. Furthermore, a comparison of the influence of different rubber particle sizes on the microstructure reveals a trend of progressive deterioration with increasing mesh size. Specimens containing 20-mesh rubber powder show better interfacial bonding between the rubber particles and the mortar than those incorporating 40-mesh and 80-mesh rubber powder. For specimens with 80-mesh rubber powder, the ITZs are more pronounced, and the number of pores is visibly higher. These defects result in greater directional discrepancies in compressive strength due to discontinuous stress transfer, leading to a significant increase in the I_A .

As shown in Fig. 12, the micro-morphology of 3D-printed rubberized concrete specimens becomes progressively more porous and less compact with increasing rubber powder replacement ratio. In the specimen with 5% replacement, the rubber particles are well bonded to the surrounding mortar, with dense paste distribution and only a few minor pores, indicating that the interfacial structure remains relatively intact at low dosage. However, as the replacement ratio increases, the volume fraction of rubber particles rises, and the hydrophobic nature of rubber leads to greater entrapped air content. This results in further microstructural degradation. In specimens with higher replacement ratios, the ITZs become noticeably wider and more porous under the same magnification. The widening of the ITZ may lead to discontinuities in stress transfer, which is considered one of the key factors contributing to the increased anisotropy in compressive strength at high rubber powder replacement levels. Moreover, Fig. 12d shows multiple micro-cracks distributed along the ITZ between rubber particles and the cement paste. This observation suggests that, under loading, crack propagation may be deflected and dispersed around the rubber–paste interface, transforming a single crack into multiple microcracks and resulting in a more tortuous damage path. The associated crack-path elongation and damage dispersion may contribute to additional energy dissipation, providing microstructural support for the potential crack-arresting effect of rubber powder in 3D-printed concrete.

3.6. Pore structure characterizes

The local microstructure of 3D-printed concrete was observed using SEM; however, it did not reveal how variations in pore size affect the compressive strength and anisotropy index. Therefore, in this section, MIP is employed to quantitatively analyze the distribution and fractal dimension of pores across different size ranges, aiming to statistically uncover the evolution of pore structure induced by rubber powder incorporation in 3D-printed concrete.

As shown in Fig. 13, both cumulative intrusion and pore size distribution curves were obtained. The results show a clear increase in total pore volume with increasing rubber content. For instance, the cumulative pore volume increased from 0.074 mL/g in the plain 3D-printed concrete to 0.082 mL/g, 0.122 mL/g, 0.097 mL/g, and 0.112 mL/g at 5%, 10%, 15%, and 20% replacement using 20-mesh rubber powder. Similar trends were observed with 40-mesh and 80-mesh particles, indicating that rubber incorporation consistently reduces mortar compactness regardless of particle size. This rise in pore volume corresponds to an increased number and extent of ITZs between rubber particles and cement paste, which introduce additional porosity and weak interfaces. In addition to the ITZ-related

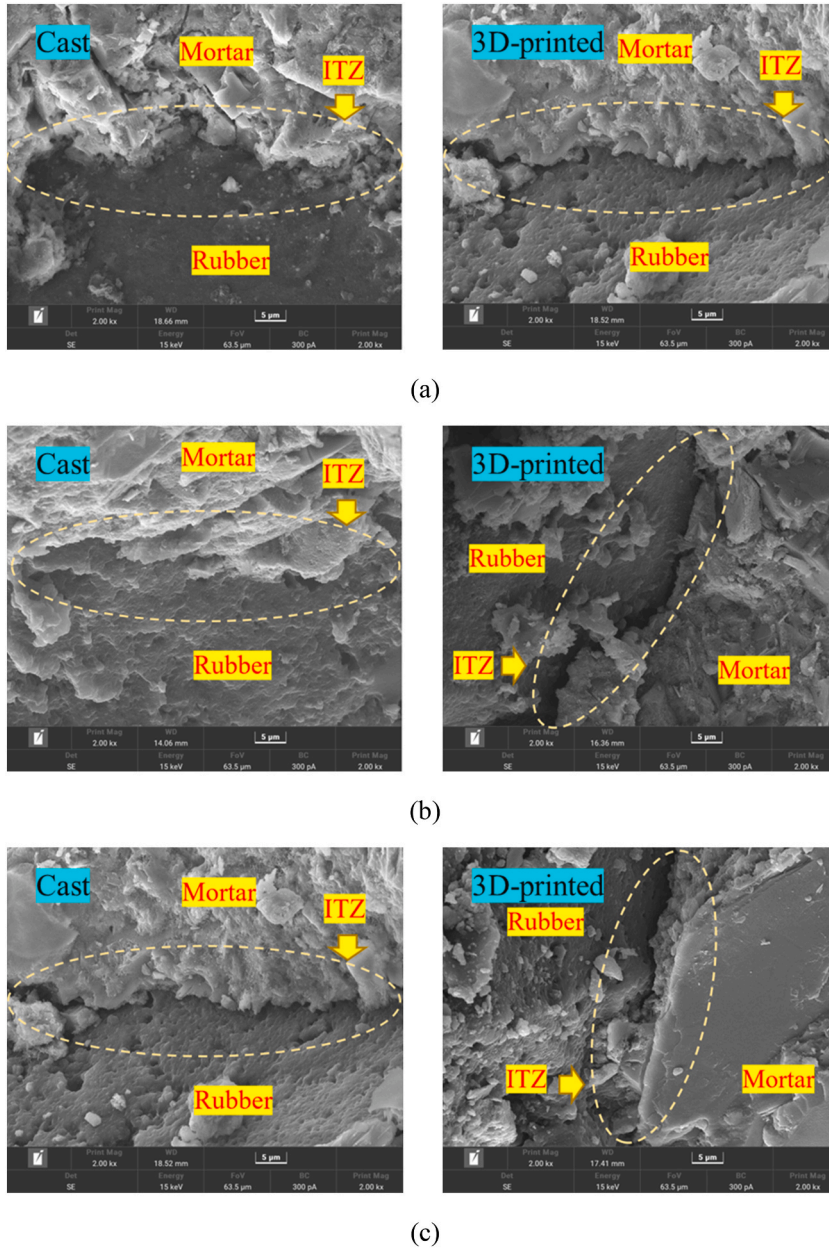


Fig. 11. Micro-morphology for cast-in-place and 3D-printed concrete specimens (a) RPC-20-5%, (b) RPC-40-5%, (c) RPC-80-5%.

porosity, the printing quality governed by fresh-state buildability can also contribute to the pore structure evolution. As discussed in Section 3.2, an increased $\tan\theta$ indicates more pronounced filament spreading and non-uniform layer thickness during layer-by-layer deposition, which may hinder effective interlayer consolidation and facilitate the formation of printing-induced pores (e.g., interlayer voids and interfilament pores). Therefore, the increase in total pore volume and the associated strength degradation/anisotropy should be understood as a combined outcome of (i) the additional ITZs introduced by rubber particles and (ii) potential buildability-related defects arising from altered fresh-state behavior. These microstructural changes significantly reduce the concrete's ability to bear and transfer compressive stress, especially in the Z-direction where the printing-induced layering is dominant. Moreover, the directional heterogeneity of pore structures and ITZ distribution leads to uneven load-bearing capacity across different orientations, thereby amplifying strength anisotropy. As confirmed by the experiments, the I_A increases from 0.288 in the plain concrete specimen to above 0.374 at 20% rubber content. Therefore, the degradation in pore structure not only accounts for the observed strength reduction but also serves as a key microstructural mechanism explaining the enhancement in compressive strength anisotropy of 3D printed rubberized concrete.

Further analysis of the log-differentiated pore size distribution curves as shown in Fig. 13 reveals that, compared to the 3D-printed

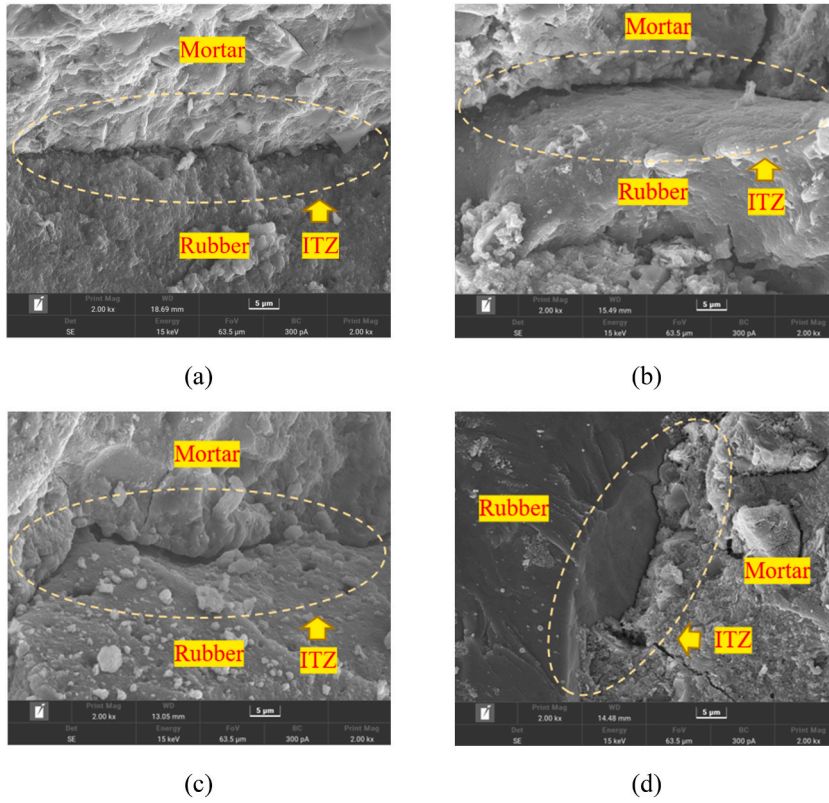


Fig. 12. Micro-morphology for 3D-printed rubber concrete with different replacement ratios (a) RPC-20-5%, (b) RPC-20-10%, (c) RPC-20-15%, (d) RPC-20-20%.

plain concrete, the incorporation of rubber powder significantly increases the most probable pore diameter, with this effect becoming more pronounced as the rubber content increases. For the 3D-printed plain concrete, the most probable pore diameter is 432.39 nm. However, after the addition of rubber powder, especially at a 20% rubber powder replacement ratio, the dominant pore diameters for the 20-mesh and 40-mesh rubber powder specimens are 551.06 nm and 551.06 nm, respectively, showing a significant increase. Results indicate that the addition of rubber powder not only increases the internal pore volume but also enhances the heterogeneity of the pore size distribution, causing the pore network to become looser and more disordered, which in turn affects the mechanical performance and increases anisotropy. The increase in pore size means a higher probability of discontinuities in the stress transfer path, further exacerbating the compressive strength anisotropy. Therefore, the incorporation of rubber powder not only causes changes in the pore structure but also leads to an increase in pore volume, reducing the efficiency of stress transfer, which is the main reason for the increased I_A in 3D-printed rubberized concrete.

3.7. Correlation between the fractal dimension and compressive strength/ I_A

The fractal dimension, a key indicator reflecting the complexity of the pore network, shows a strong correlation with the mechanical behavior of concrete. Typically, a more intricate pore structure in concrete results in a wider pore size distribution and a higher fractal dimension value. As the fractal dimension increases, the pore morphology becomes more irregular, and more weak interfaces form within the concrete. The increase in these weak interfaces is one of the main reasons for the loosening of the concrete structure and the deterioration of its mechanical performance. For 3D-printed concrete specimens, existing studies have established a power function model to describe the correlation between fractal dimension and compressive strength, demonstrating a high degree of fit. The pore fractal dimension D is calculated as follows [58]:

$$\ln \frac{W_n}{r_n^2} = D \ln Q'_n + \ln C \quad (5)$$

where, W_n and Q'_n are the shorthand for $\sum_{i=1}^n P_i \Delta V_i$ and $\frac{1}{r_n^3}$ respectively; P_i is the pressure at the i -th mercury intrusion, Pa; ΔV_i presents the intrusion volume at the i -th intrusion, in m^3 ; n is the number of mercury intrusion; r_n is the pore radius at the n -th intrusion, in meters (m); V_n is the cumulative intrusion volume, in m^3 ; C is a regression constant; D is the fractal dimension of the pore surface.

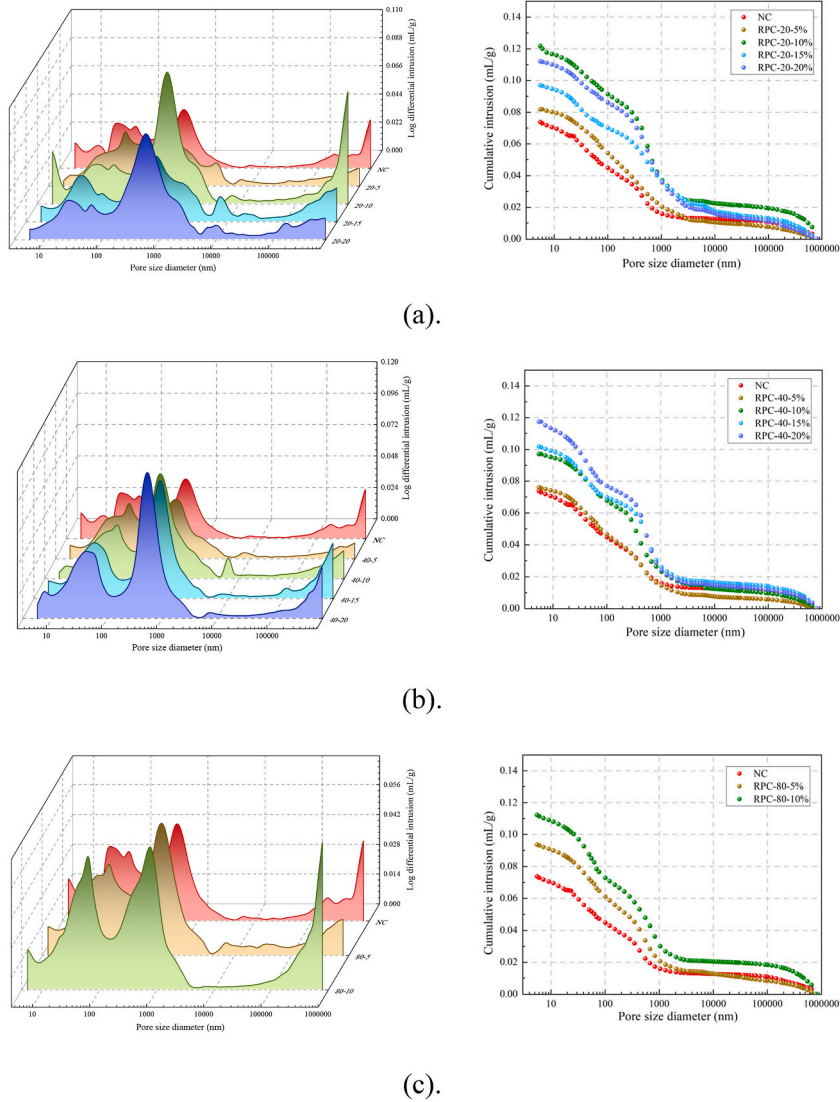
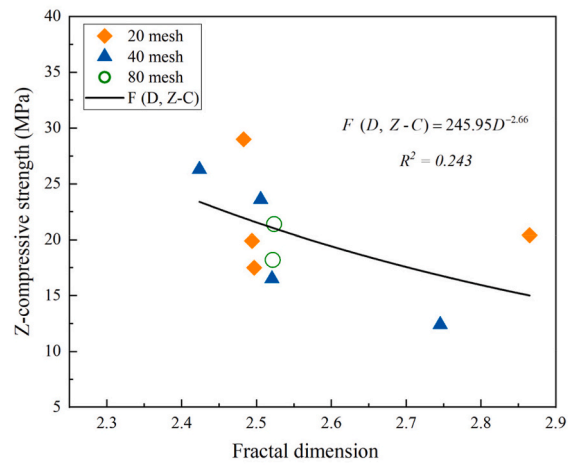


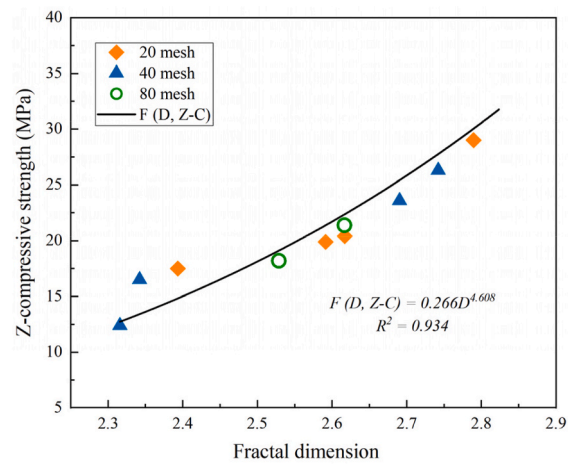
Fig. 13. Effect of rubber powder size and content on the pore size distribution (a) rubber powder size of 20-mesh; (b) rubber powder size of 40-mesh; (c) rubber powder size of 80-mesh.

According to Eq. (5), $\frac{W_n}{r_n^2}$ and Q_n can be calculated from the data of the MIP test. Then, the fractal dimension D can be calculated by the $\frac{W_n}{r_n^2}$ and Q_n . Following Visser et al., [59,60] pores are classified by diameter into micropores (<10 nm), capillary pores (10–200 nm), and macropores (>200 nm). According to Jin et al. [58], among various pore-related parameters such as porosity and fractal dimensions of different pore types, the fractal dimension demonstrates a stronger power-law relationship with compressive strength. Notably, the fractal dimension of capillary pores exhibited the highest correlation coefficient (R^2) of 0.862, indicating its critical role in predicting mechanical performance. The SEM and MIP results indicate that incorporating rubber powder produces a pore-structures distinct from that of 3D-printed plain concrete: as the replacement ratio increases, overall porosity rises and the pore-size spectrum shifts, which in turn affects strength. Therefore, it is crucial to further explore the correlation between strength and pore distribution in 3D-printed rubberized concrete. Understanding the relationship between fractal dimensions of different pore types and the material's strength will be instrumental in optimizing mix designs. Specifically, the link between the I_A and the fractal dimension is vital for enhancing the mechanical performance and compressive strength of 3D-printed rubberized concrete.

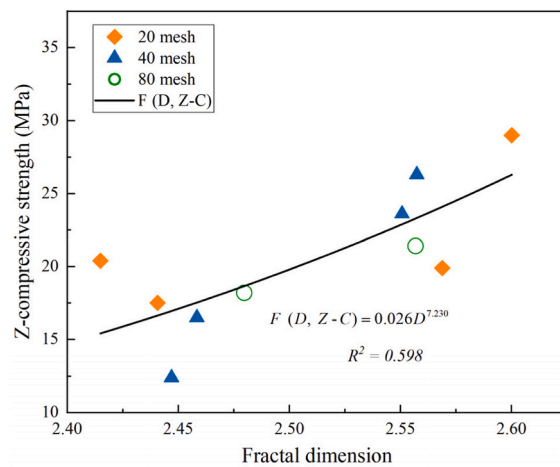
Fig. 14 illustrates the relationship between compressive strength in the FZ direction and the fractal dimensions of macropores, capillary pores, and micropores. As illustrated in Fig. 14a, a higher fractal dimension of micropores corresponds to a decrease in compressive strength in the FZ direction. This trend suggests that a higher micropore fractal dimension, reflecting a more complex and heterogeneous micropore network, may hinder the continuity of stress transfer and thereby reduce compressive strength. By contrast, within the investigated range, the fitted trends for capillary pores and macropores in Fig. 14b and c shows positive correlations with



(a).



(b).



(c).

Fig. 14. Fractal dimension vs. compressive strength in the FZ direction. (a). micropores; (b). capillary pores; (c). macropores.

the FZ compressive strength. Within the investigated range (20-80 mesh and 5%-20% replacement ratio), the power-law empirical correlation based on the capillary-pore fractal dimension achieves a goodness-of-fit of $R^2 = 0.93$. In contrast, the fractal dimensions of other pore types exhibited relatively low R^2 values when correlated with compressive strength. These results indicate that, within the present dataset, the capillary-pore fractal descriptor provides the most effective statistical indicator for FZ compressive strength among the examined pore ranges. This aligns with Jin et al. [58], who found that in cast mortar, the fractal dimension of capillary pores had a stronger correlation with compressive strength in the FZ direction than other pore types, establishing a power-law relationship. However, further validation is needed to confirm the model's applicability beyond this range, especially at higher contents or finer particle sizes.

Fig. 15 presents the empirical correlations between the I_A and pore fractal dimensions for 3D-printed rubberized concrete within the investigated range. The results indicate that the I_A increases with the fractal dimension of micropores, which suggests that the increase in micropores and the complexity of the pore structure led to greater heterogeneity in the pore structure. This, in turn, affects the mechanical properties of the material, particularly the uneven stress transfer across different directions, thereby increasing compressive strength anisotropy. As the fractal dimension of micropores increases, the distribution of pores becomes more irregular and complex, preventing uniform stress transfer and further exacerbating the rise in I_A . In contrast, the anisotropy index decreases with the increase in the fractal dimension of capillary and macropores. As the fractal dimension increases, the pore structure becomes more ordered, and the effective connectivity between pores enhances stress transfer. This results in a decrease in I_A , indicating reduced compressive strength anisotropy. Notably, the power-law fit between I_A and macropore fractal dimension shows the highest goodness-of-fit within the investigated range (20-80 mesh and 5%-20% replacement ratio), with $R^2 = 0.83$. Unlike the compressive strength in the FZ direction, the I_A exhibits a stronger correlation with the fractal dimension of macropores. This may be related to macropore-dominated defects in extrusion-based printing (e.g., entrapped air voids and interlayer/interfilament pores) and the limited compaction inherent to the deposition process, which can enlarge directional differences in load transfer and thereby increase I_A .

4. Conclusions

This work examined how rubber powder particle size and dosage affect the buildability of fresh 3D-printed concrete and the compressive behavior of hardened specimens under various loading directions. To further elucidate the micro-macro connection, MIP and SEM were used to characterize the pore system and microstructure. Moreover, the correlations among pore fractal dimensions, compressive strength, and I_A were analyzed. The main findings are summarized as follows.

- The addition of rubber powder improves the fluidity of the concrete, which generally benefits the extrusion process in 3D printing. However, when the rubber powder is ground to finer sizes (e.g., 80 mesh) and the replacement ratio exceeds 10%, the excessive fluidity may compromise buildability and lead to structural instability or printing failure.
- Reducing the particle size of rubber powder exerts a positive effect on enhancing the buildability of rubberized concrete, thereby achieving a balance between fluidity and printing stability.
- Compared to cast concrete, the compressive strength in the vertical direction is significantly diminished due to the weak interfaces between the printed layers and filaments. MIP results further reveal that, in this orientation, the fractal dimension of capillary pores shows a stronger power-law relationship with compressive performance.
- At a 5% rubber powder replacement ratio, the compressive strength anisotropy index of 3D-printed concrete can be effectively reduced. However, at elevated replacement ratios, the anisotropy index tends to rise. The fractal dimension of macropores shows the best power-function relationship with the anisotropy index, indicating its dominant influence on strength anisotropy.

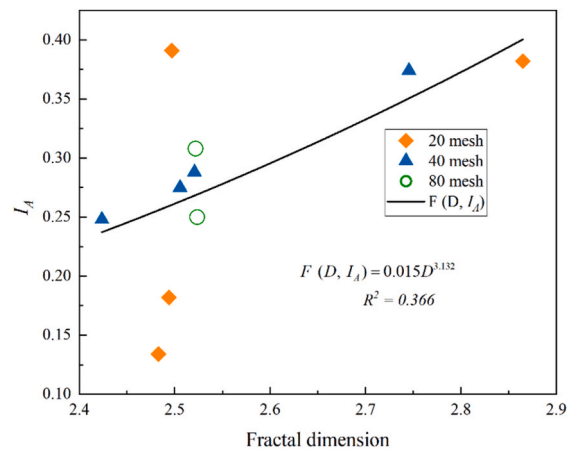
5. Future scopes

In the present study, the authors primarily focused on the static performance of 3D-printed concrete, investigating the effects of rubber powder particle size and replacement ratio on buildability, compressive strength anisotropy, and pore-structure characteristics. It should be noted that the toughening and energy-dissipation effects of rubber incorporation in 3D-printed concrete were not further addressed in this work. For example, damping-related behavior and quantitative toughness/energy-absorption evaluation of 3D-printed rubberized concrete specimens still warrant systematic investigation in future studies. In addition, as discussed in the Introduction, fibers are also commonly used to enhance the toughness of 3D-printed concrete. Therefore, a direct comparison between rubber and fibers under identical printing conditions is needed to further assess their relative toughening efficiency and applicability.

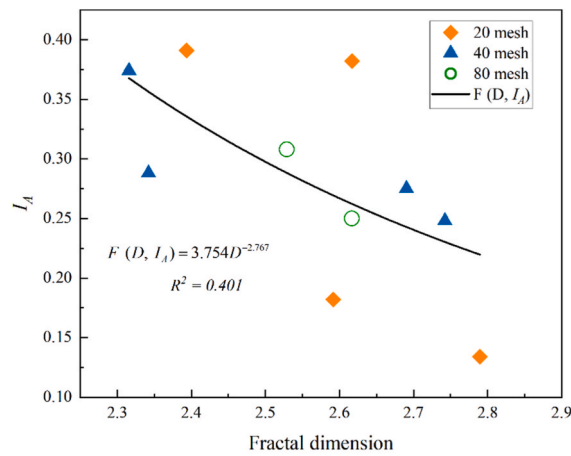
Although the proposed power-law correlations between fractal dimension and compressive strength/ I_A show good fit within the investigated window (20-80 mesh and 5%-20% replacement ratio), their transferability to broader rubber parameters and different printing conditions requires further verification. Future studies will expand the dataset to include wider rubber dosages and finer particle sizes, and will examine the robustness of model coefficients under varied printing parameters, with the aim of establishing a more generally applicable microstructure-informed prediction framework for anisotropy in extrusion-based printed cementitious composites.

CRedit authorship contribution statement

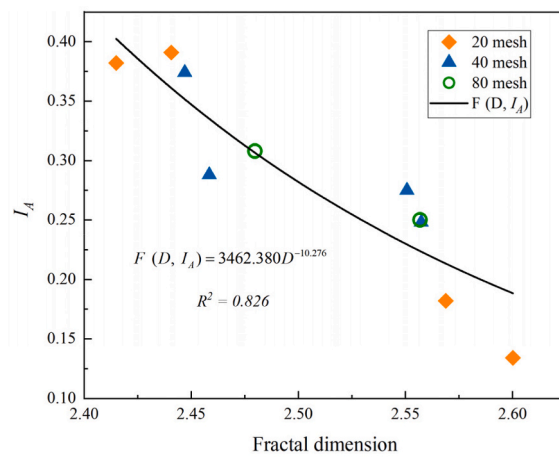
Jinxu Mo: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation. **Han Lu:** Methodology, Investigation. **Fengming Ren:** Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition,



(a).



(b).



(c).

Fig. 15. Fractal dimension vs. I_A . (a). micropores; (b). capillary pores; (c). macropores.

Conceptualization. **Mianheng Lai**: Methodology, Investigation. **Johnny Ching Ming Ho**: Methodology. **Shaozu Guo**: Methodology, Investigation. **Ruiming Sun**: Methodology. **Shiyu Tian**: Investigation. **Jianrong Xiong**: Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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